# Algebra 2
## Practice Test

# Unit 1: Linear Functions

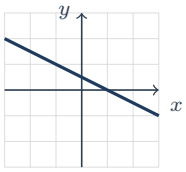
**Instructions:** Show all work. Write every line in $y = mx + b$ form when asked. No notes or calculator unless your teacher says so. Write each final answer in its box. Only what is inside a box is graded. **Total: 100 pts**

**Keep in Mind**

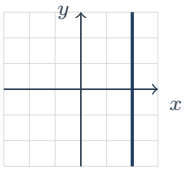
**This is a practice test.** It mirrors the real Unit 1 test in length, format, and the ideas it covers, but the numbers differ. Work every problem with no notes, then check your answers against the key.

## Part A — Types of Slope (8 pts)

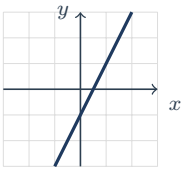
For each line, write whether its slope is **positive**, **negative**, **zero**, or **undefined**.



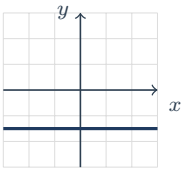

1. ☐




2. ☐




3. ☐




4. ☐

No graph this time — decide from the points or the equation.

5. The line through $(3, 2)$ and $(-1, 2)$: ☐

6. The line $y = 5x - 1$: ☐

7. The line $x = -4$: ☐

8. The line through $(0, 6)$ and $(2, 1)$: ☐

## Part B — Writing Equations of Lines (20 pts)

Write each line in slope-intercept form, $y = mx + b$. Show how you found $m$ and $b$; for 11 and 12, start in point-slope form, $y - y_1 = m(x - x_1)$.

9. The line through $(1, 5)$ and $(3, 9)$.

Equation: ☐

10. The line through $(-4, 5)$ and $(2, -4)$.

Equation: ☐

11. The line with slope 2 that passes through $(3, 4)$.

Equation: ☐

12. The line with slope $-\frac{1}{3}$ that passes through $(6, 1)$.

Equation: ☐

## Part C — Piecewise Functions (20 pts)

13. **Evaluate.** (8 pts) $f(x) = \begin{cases} 3x + 2, & x < -1 \\ -1, & -1 \le x \le 1 \\ 2x - 5, & x > 1 \end{cases}$

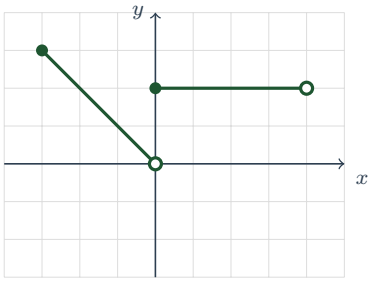
$f(-2) =$ ☐ $f(-1) =$ ☐ $f(1) =$ ☐ $f(4) =$ ☐

14. **Domain.** (12 pts) Give the domain of each function as an inequality or in interval notation.

(a) $g(x) = \begin{cases} x - 2, & x < 3 \\ 7, & 3 \le x < 6 \end{cases}$ Domain: ☐

(b) $h(x) = \begin{cases} x + 1, & -2 \le x \le 0 \\ 4 - x, & 0 < x \le 5 \end{cases}$ Domain: ☐

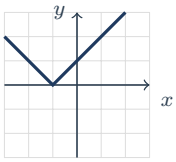
(c) The function $k$ graphed below.



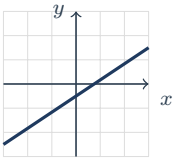

Domain: ____

## Part D — The Absolute Value Function (28 pts)

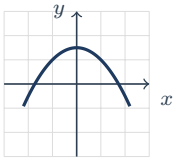
15. (3 pts) Write the letter of the graph of an **absolute value** function in its Letter box. **Letter:** ____



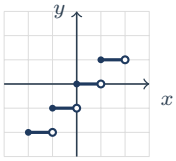

(A)




(B)




(C)




(D)

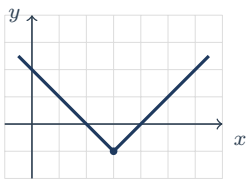
16. (3 pts) Which equation is an **absolute value** function? **Letter:** ____

(A) $y = x^2 - 3$ (B) $y = 2x + 5$ (C) $y = |x + 1| - 3$ (D) $y = \lfloor x \rfloor + 1$

17. (15 pts) Each function is a transformation of the parent $y = |x|$. Fill in the table.

| Function | Vertex | Left/right shift | Up/down shift | Opens |
|---|---|---|---|---|
| $g(x) = \|x + 2\| + 3$ | | | | |
| $h(x) = \|x - 5\| - 4$ | | | | |
| $k(x) = -\|x\| - 2$ | | | | |

18. (4 pts) Which equation matches the graph? **Letter:** ____




(A) $y = |x + 3| - 1$
(B) $y = |x - 3| + 1$
(C) $y = |x - 3| - 1$
(D) $y = -|x - 3| - 1$

19. (3 pts) Compared with the parent $y = |x|$, the graph of $y = \frac{1}{2}|x|$ is **Letter:** ☐

(A) narrower (B) wider (C) shifted down 2 (D) shifted left 2

## Part E — Absolute Value Equations & Inequalities (24 pts)

Solve. Show both cases.

20. (8 pts) $|2x + 1| = 9$

Solution:

21. (8 pts) $|x - 3| \leq 2$

Solution:

22. (8 pts) $|x + 4| > 6$

Solution: